



Department of Computer Applications
CIE-I Second Semester MCA

Subject Name: Elective I (Data Communication and Computer Networks)

Subject Code: 25MCA25X

Time: 1 hr

Max. Marks: 20

Course Outcomes:

CO1: Explore the basic concepts, principles, and techniques of data Communication along with the layers of OSI and TCP/IP model.
CO2: Discuss various error detection and correction techniques in data link layer and analyze the performance of network layer.
CO3: Demonstrate the services of transport layer and the basic concepts of Application Layer Paradigms.
CO4: Explore the fundamental concepts of Cryptography by applying various encryption techniques
CO5: Analyze various asymmetric ciphers and data integrity algorithms.

PART A: ANSWER ALL THE QUESTIONS

Each question carries 1 mark 4Qx1=04

Q	QUESTIONS	BL	CO
1	a State the four basic components of a data communication system.	1	1
	b Describe simplex and full-duplex modes of data transmission	2	1
	c Define bandwidth and bit rate in communication systems.	1	2
	d What is the main purpose of protocol layering in computer networks?	2	2

PART B: ANSWER ALL THE QUESTIONS

Each question carries 4 marks 3Qx4=12

Q	QUESTIONS	BL	CO
2	a Explain how multiplexing and demultiplexing are used in network communication.	2	1
	OR		
	b Describe the types of addressing used at different layers of the TCP/IP model.	2	1
3	a Differentiate between LAN, MAN, and WAN based on their key characteristics	3	1
	OR		
	b Compare the OSI model and the TCP/IP model in terms of architecture and functionality.	2	1
4	a Illustrate the working of different switching techniques using suitable examples.	3	2
	OR		
	b Define transmission impairment. A communication channel experiences noise and attenuation during transmission. Analyze how these impairments affect data communication and suggest ways to improve signal quality.	4	2